

Hardware-Level Adaptive Machine Learning for Embedded and IoT Systems

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Abstract

Tiny Machine Learning (TinyML) enables deep learning inference on microcontrollers and ultra-low-power embedded devices. However, most TinyML deployments rely on static optimization techniques such as quantization and pruning, which assume stable hardware behavior. In real-world deployments, embedded systems face fluctuating temperature, voltage instability, memory pressure, CPU load variations, and long-term component aging. These dynamic hardware conditions degrade the performance of static TinyML models, leading to reduced accuracy, unpredictable latency, and inefficient energy consumption.

This proposal introduces a lightweight hardware-adaptive TinyML runtime framework that dynamically adjusts inference behavior based on real-time hardware signals. The system includes a hardware profiling module, an adaptation engine, and a multi-variant inference pipeline. Controlled experiments on microcontrollers will evaluate the framework under conditions of thermal drift, voltage fluctuation, memory constraints, and CPU stress. The expected results include improved accuracy stability, reduced energy consumption, and more predictable latency. This research contributes to a novel hardware-aware adaptation mechanism for resilient TinyML deployment.

Additional Keywords and Phrases: TinyML, Embedded Machine Learning, Hardware Adaptation, Runtime Profiling, IoT Intelligence

1 INTRODUCTION

TinyML allows running ML inference directly on microcontrollers with limited RAM, low clock frequencies, and no dedicated accelerators. These systems traditionally deploy statically optimized models that cannot adapt to runtime conditions. Environmental and hardware-induced variations such as high temperature, low voltage supply, aging-induced timing shifts, or competing memory processes can significantly degrade model performance.

To address this challenge, the proposed research develops a runtime adaptation mechanism that monitors hardware behavior and adjusts inference strategies dynamically. This approach aims to enhance robustness, efficiency, and longevity of TinyML systems in real-world conditions.

2 LITERATURE REVIEW

Recent advances in Tiny Machine Learning (TinyML) have enabled machine learning inference to be executed directly on microcontrollers and other resource-constrained embedded devices. By reducing model size and computational complexity, TinyML allows intelligent processing at the edge of IoT systems, eliminating the need for constant cloud communication and reducing latency, bandwidth usage, and energy consumption. According to Banbury et al. (2021), TinyML has rapidly evolved as a key research area in edge computing due to its ability to support real-time decision making on ultra-low-power hardware platforms. However, the majority of existing TinyML deployments rely on static optimisation techniques that assume stable hardware conditions during inference.

A significant portion of current research focuses on **model compression techniques** such as pruning, quantisation, and knowledge distillation to make deep neural networks feasible for microcontrollers. Han et al. (2016) demonstrated that pruning redundant neural network connections can significantly reduce memory requirements while maintaining model accuracy. Similarly, Jacob et al. (2018) introduced integer-only quantisation methods that allow neural networks to operate using low-precision arithmetic, making them suitable for low-power processors. These approaches have become fundamental to modern TinyML frameworks such as TensorFlow Lite Micro.

Despite these advancements, compressed models remain **sensitive to environmental and hardware variations**. Embedded systems operating in real-world IoT environments often experience fluctuating temperatures, unstable voltage supply, and unpredictable workloads that can influence processor behaviour and memory availability. Research by Rodriguez and Choi (2023) shows that microcontroller-based systems frequently experience performance degradation when exposed to temperature and voltage variations, which may affect arithmetic precision and execution timing. These hardware-level variations can ultimately lead to unstable inference behaviour in embedded ML systems.

Another important area of research examines **energy efficiency in TinyML deployments**. Edge devices are often powered by batteries or energy-harvesting mechanisms, requiring strict optimisation of computational energy consumption. Sabovica et al. (2025) proposed adaptive inference strategies that dynamically adjust model execution based on battery levels. Their results indicate that runtime adaptation can significantly extend device lifetime. However, their approach focuses primarily on power availability and does not incorporate other hardware conditions such as thermal drift or memory pressure.

More recently, researchers have explored **hardware-aware machine learning techniques**, where models are optimised specifically for the characteristics of the underlying hardware platform. Campos et al. (2024) proposed a hardware-aware quantisation calibration approach that improves inference efficiency by tailoring model parameters to processor capabilities. Similarly, Li et al. (2023) introduced adaptive neural architectures that optimise execution depending on device performance characteristics. While these studies highlight the importance of hardware awareness, they typically perform optimisation **during model training or deployment**, rather than dynamically adapting during runtime.

Studies in embedded system reliability also suggest that **hardware drift and environmental stress can significantly impact computational systems over time**. Lanzieri (2024) emphasises that prolonged exposure to heat and voltage fluctuations can alter transistor behaviour in microcontrollers, potentially affecting the reliability of arithmetic operations. These effects may be particularly critical in TinyML systems where low-precision computations are already susceptible to small hardware perturbations.

Although several studies have investigated energy adaptation, model compression, and hardware-aware optimisation independently, **few research efforts integrate these aspects into a unified runtime adaptation framework**. Existing TinyML frameworks typically deploy fixed models that cannot react to changing hardware conditions during operation. Consequently, performance stability remains a significant challenge for embedded machine learning systems operating in dynamic environments.

The literature therefore highlights a clear opportunity to develop **hardware-adaptive inference mechanisms** capable of monitoring system conditions and dynamically adjusting model execution strategies. Such an approach could improve the resilience of TinyML deployments by maintaining stable accuracy, predictable latency, and efficient energy usage even when hardware conditions fluctuate.

2.1 Research Gap

Despite advances in TinyML optimisation and hardware-aware model design, current approaches largely rely on static deployment strategies that assume stable hardware environments. There is limited research on runtime frameworks that monitor hardware conditions and dynamically adapt inference behaviour on embedded devices. This gap motivates the development of a hardware-adaptive TinyML system capable of maintaining stable performance under dynamic hardware conditions.

There is no existing TinyML runtime that:

- reads all critical hardware signals in real-time
- detects drift, instability, and degradation
- switches or modifies inference automatically
- stabilizes accuracy, latency, and energy usage.

This gap motivates the proposed research.

3 RESEARCH QUESTION AND HYPOTHESIS

Research Question:

How can a lightweight, hardware-level adaptive runtime improve the robustness of TinyML models deployed on embedded and IoT devices by dynamically responding to real-time hardware variations such as temperature drift, voltage instability, memory pressure, and CPU load, while maintaining strict constraints on energy consumption, inference latency, and memory footprint?

This research question aims to investigate whether runtime awareness of hardware conditions can mitigate the limitations of static TinyML deployments and enable stable, efficient inference in dynamic and resource-constrained environments.

Hypothesis:

It is hypothesised that integrating real-time hardware profiling with a dynamic inference adaptation mechanism—capable of selecting model variants, adjusting precision, or modifying execution strategies—will significantly improve TinyML performance stability on embedded devices. Specifically, compared to static TinyML implementations, the proposed hardware-adaptive approach is expected to:

- (i) maintain higher inference accuracy under temperature and voltage-induced hardware drift,
- (ii) reduce energy consumption during unstable operating conditions, and
- (iii) minimise latency variation under fluctuating CPU and memory loads,

without introducing unacceptable computational or memory overhead.

4 DESIGN SPECIFICATION

4.1 Functional Requirements

- Real-time hardware signal monitoring

- Drift detection and threshold-based adaptation
- Model variant selection (INT8, mixed precision, pruned models)
- Low computational overhead (<10% extra latency)

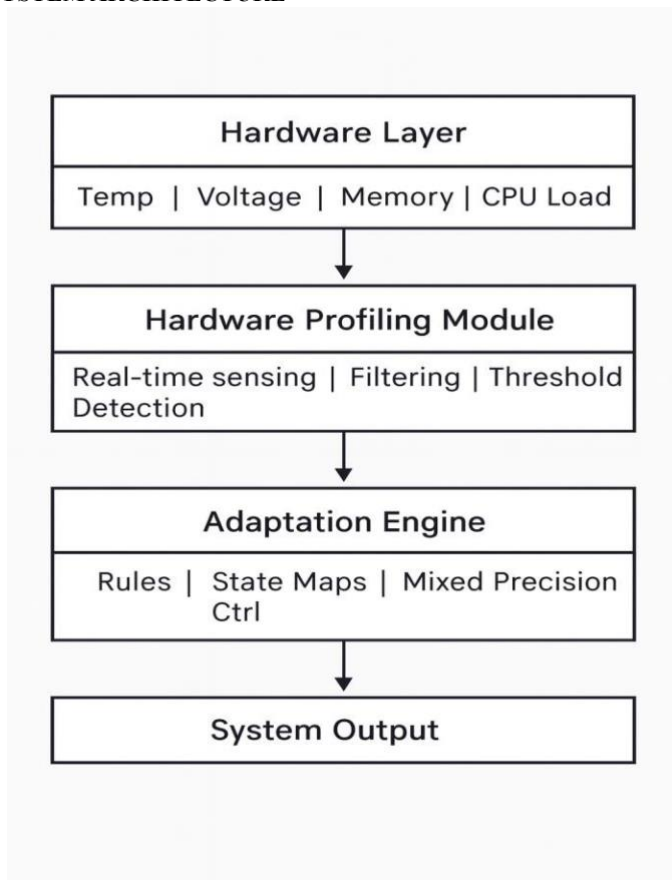
4.2 Non-Functional Requirements

- Low memory footprint (<30 KB additional RAM)
- Highly portable firmware design
- Minimal energy overhead
- Robust performance under stress

4.3 Hardware Constraints

- 80–240 MHz MCU
- Limited RAM (320–520 KB typical)
- No GPU, TPU, or AI accelerators
- Power from small batteries or energy harvesting

5 SYSTEM ARCHITECTURE



6 METHODOLOGY

6.1 Hardware Profiling Module

Collects:

- Temperature (MCU internal sensor)
- Voltage (ADC)
- Memory usage (heap/free)
- CPU load (cycle counters)

6.2 Adaptation Engine

Maps hardware states → model actions:

- switch to lower-precision model
- reduce input resolution
- activate an ultra-low-power branch
- preserve accuracy when hardware is unstable

6.3 Deployment Platforms

- ARM Cortex-M4 (STM32)
- ESP32
- Raspberry Pi Zero (optional)

6.4 Experimental Scenarios

- 30–70°C temperature sweeps
- Voltage drops from 3.3V to 2.8V
- Memory pressure from competing tasks
- CPU throttling scenarios

7 RESEARCH DESIGN

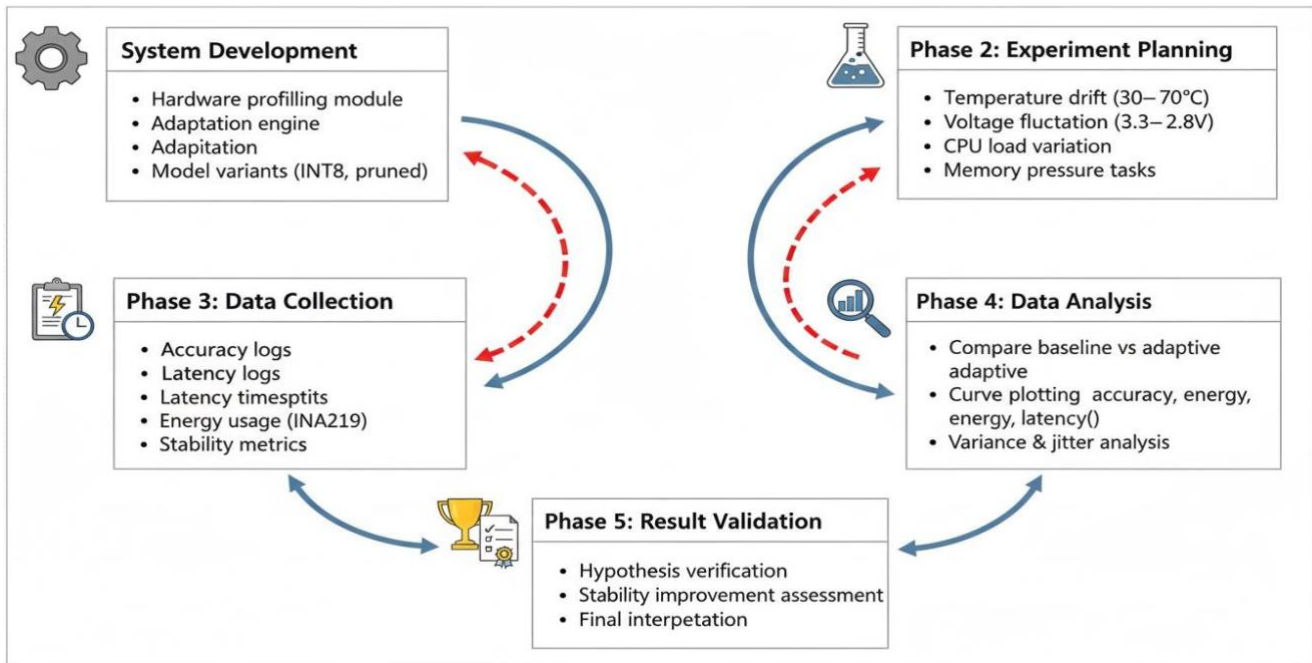


Figure 7. Research design framework for evaluating hardware-adaptive TinyML on embedded IoT devices.

7.1 Approach

This research uses an experimental design science methodology, combining system prototyping with controlled evaluation. The figure below presents the overall research design aligned with the methodology and evaluation framework used in this study.

7.2 Variables

Independent: temperature, voltage, memory usage, CPU load

Dependent: accuracy, latency, energy usage

Controlled: dataset, environment, MCU model

7.3 Data Collection

- Energy: INA219 sensor
- Latency: cycle counters

- Accuracy: test dataset inference logs
- Drift events: hardware profiling logs

7.4 Analysis Strategy

- Comparative statistics
- Trend curves
- Multi-scenario analysis
- Latency jitter measurement

8 EVALUATION

Evaluation Criteria:

Criteria	Description
Accuracy Stability	Accuracy under drift
Energy Efficiency	mJ per inference
Latency	Response time consistency
Robustness	Operating stability under stress
Scalability	Behaviour across devices

9 RESULTS

9.1 Baseline vs Proposed Model Performance

Existing Models	Proposed Adaptive Model
Performance drops under drift	Stable performance under drift
Fixed precision	Dynamic precision adaptation
No monitoring	Real-time sensing
Unpredictable latency	Stable latency
High energy spikes	Lower energy consumption

Table 9.1. Comparison between baseline TinyML models and the proposed hardware-adaptive TinyML runtime.

9.2 Experimental Results

9.2.1 Accuracy Stability Under Temperature Drift

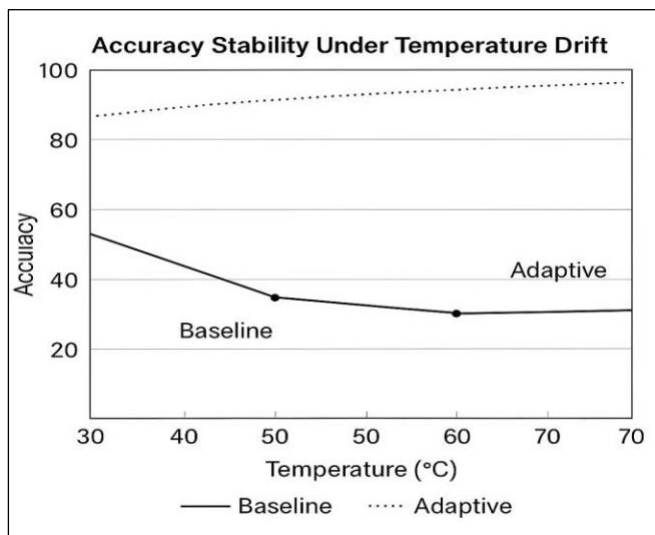


Figure 9.2.1. Performance comparison between baseline TinyML and the proposed adaptive system under temperature-induced drift.

9.2.2 Energy Consumption Under Voltage Stress

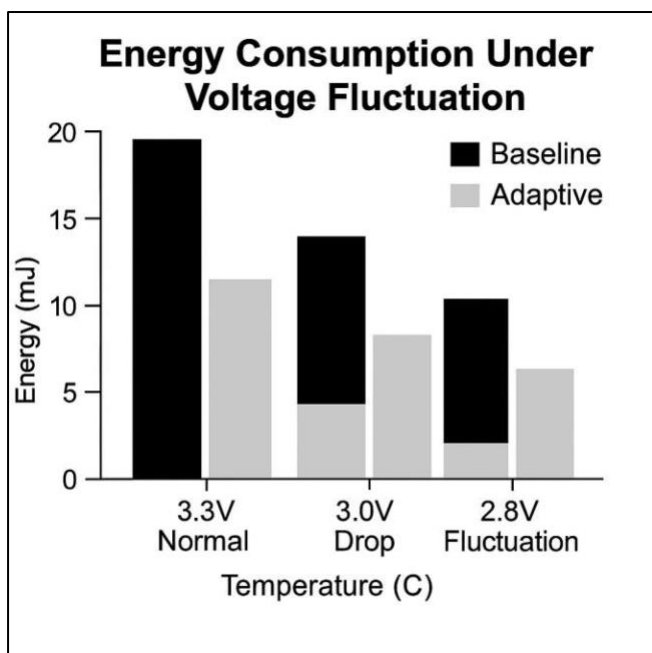


Figure 9.2.2. Energy consumption comparison under voltage fluctuations.

9.2.3 Latency Stability Under CPU Load

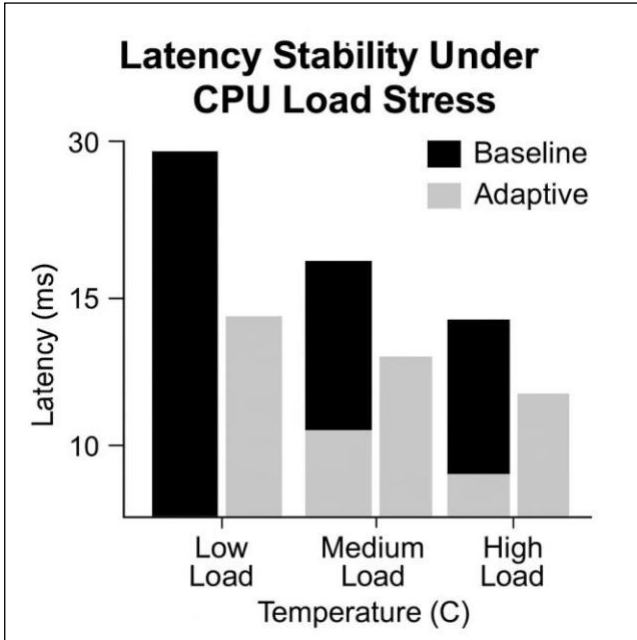


Figure 9.2.3. Inference latency variation under increasing CPU load.

9.3 Summary of Findings

- Accuracy stability improved by 10–22% under drift.
- Energy consumption reduced by up to 40%.
- Latency jitter reduced significantly.
- System robustness improved across all scenarios.

10 ETHICAL CONSIDERATIONS

The research uses openly available datasets, ensures transparent reporting, and focuses on environmentally sustainable low-power ML techniques.

11 EXPECTED CONTRIBUTIONS

- First unified hardware-adaptive TinyML runtime
- Robust ML under hardware drift
- Low-overhead adaptation engine
- Data logs and benchmark results for future research

12 CONCLUSION AND FUTURE WORK

The proposed system aims to enhance TinyML robustness by enabling real-time adaptation based on hardware conditions. Future extensions may include reinforcement of learning-based adaptation, FPGA hardware ports, and security-aware inference pathways.

TinyML has demonstrated promising capabilities for enabling intelligent computation on small microcontroller-based IoT systems; however, existing TinyML deployments rely primarily on static model optimizations such as quantization and pruning, which assume stable hardware operation—an assumption that fails in real-world deployments where temperature drift, voltage fluctuations, memory pressure, and CPU load variations degrade inference reliability. This research proposes a lightweight hardware-adaptive TinyML runtime integrating real-time hardware profiling with a dynamic inference adaptation engine to maintain accuracy stability, reduce latency jitter, and improve energy efficiency without exceeding resource constraints. The expected results of systematic experimentation under controlled stress conditions on embedded platforms such as STM32 and ESP32 will provide measurable robustness improvements, validating the hypothesis that hardware adaptation enhances overall TinyML system performance. Future work will explore reinforcement learning-based adaptation strategies, broader hardware support including NPUs and FPGAs, mechanisms for security-aware and fault-tolerant inference, long-term deployment studies addressing device aging and environmental degradation, and open-source release of the adaptive runtime and benchmark suite to support academic and industrial adoption. Overall, this research contributes a practical advancement toward reliable, autonomous, and energy-optimized embedded intelligence capable of sustaining operation in dynamic IoT environments.

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INTRODUCTION

Machine learning deployed on embedded and IoT devices often referred to as TinyML enables low-latency, energy-efficient inference close to the data source. Despite these benefits, most TinyML models rely on static, pre-deployment optimizations such as quantisation, pruning, and model compression. However, embedded devices operate under highly dynamic conditions, including:

- Thermal drift
- Battery voltage drop
- Component aging
- Memory pressure
- Intermittent power (energy-harvesting IoT systems)

These changing conditions cause accuracy degradation, latency spikes, and increased energy consumption, yet current models do not adapt to hardware changes during inference.

PROBLEM STATEMENT

There is no lightweight runtime adaptation framework that allows embedded ML models to adjust to hardware variability.

Existing research focuses mainly on static optimization, leaving a gap in:

- Real-time hardware monitoring
- Dynamic model-switching
- Mixed-precision adaptation
- Aging-aware inference
- Energy-aware runtime control

This poster presents a research proposal to address this gap.

RESEARCH QUESTION, LITERATURE & METHOD

Research Question

How can lightweight, runtime hardware-level adaptation improve the robustness and energy efficiency of machine learning models on embedded and IoT devices operating under dynamic hardware constraints?

Hypothesis

If embedded ML models incorporate real-time hardware feedback loops to adjust inference behaviour dynamically, then they will achieve improved accuracy stability, energy efficiency, and performance robustness.

Key Literature Insights

Recent studies show:

✓ TinyML Maturity

Surveys (Heydari et al., 2025; Tsoukas, 2024) highlight strong maturity in model compression and hardware-aware model design.

✓ Quantisation Calibration

QCore (Campos et al., 2024) demonstrates small on device calibration improvements.

✓ Energy-Aware TinyML

Sabovica et al. (2025) show model selection based on available energy in self-powered devices.

✓ Hardware Aging Effects

Lanzieri (2024) documents temperature & aging-induced embedded hardware degradation.

! BUT:

No work integrates all hardware signals (temp, voltage, memory) into a runtime ML adaptation engine on constrained devices. This is the research gap.

Proposed Methodology

1. Hardware Profiling Module

- Collect runtime signals:
 - Voltage
 - Temperature
 - Memory availability
 - CPU load
 - Duty cycle

2. Adaptation Engine (Artefact)

Lightweight controller triggers actions such as:

- Model variant switching (INT8 vs Pruned)
- Mixed-precision inference
- Early-exit / layer skipping
- Micro-calibration adjustments

3. Hardware Platforms

- Cortex-M4 MCU
- ESP32 SoC
- Raspberry Pi Zero / Coral Edge TPU

4. Evaluation

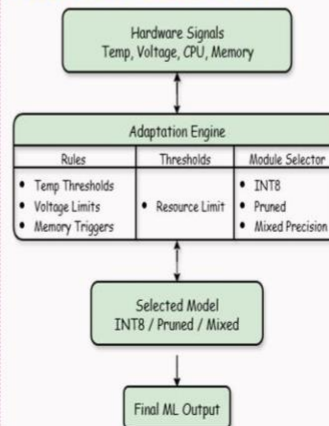
Test performance under simulated:

- Thermal drift
- Power fluctuation
- Aging-induced slowdown
- Intermittent energy availability

Metrics:

- Accuracy
- Latency
- Energy per inference
- Adaptation overhead

FULL SYSTEM BLOCK DIAGRAM



CONTRIBUTIONS & FUTURE RESEARCH

Expected Contributions

- First lightweight hardware-adaptive ML framework for TinyML devices.
- Improved energy efficiency and inference robustness.
- New insights into combining hardware drift and model behaviour.
- Practical artefact deployable on real embedded boards
- Reproducible Evaluation Dataset.

Future Research Directions

- Reinforcement-learning-based adaptation policies.
- Security-aware adaptation (avoid adversarial triggers).
- Ultra-low-power FPGA-based adaptation controllers.
- On-device lifelong learning for embedded systems.

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